

DIGITAL LOGIC DESIGN

(CE_118)

CHAPTER 2:

Storage Components

(part_1)

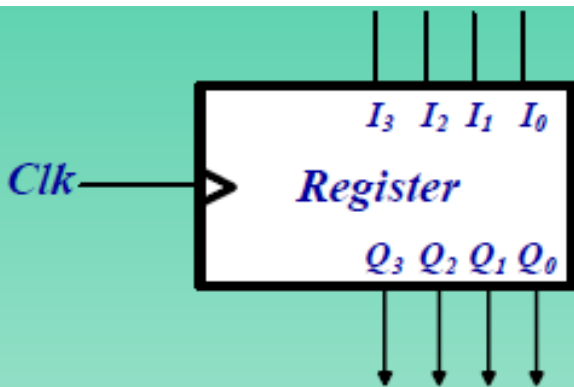
Storage components

- Storage components store data and perform some simple operations.
- Storage components include:
 - registers
 - counters
 - register files
 - queues
 - stacks
- Combinatorial and storage components are used for construction of:
 - datapaths
 - controllers
- Main subsystems of modern processors and other microchips.

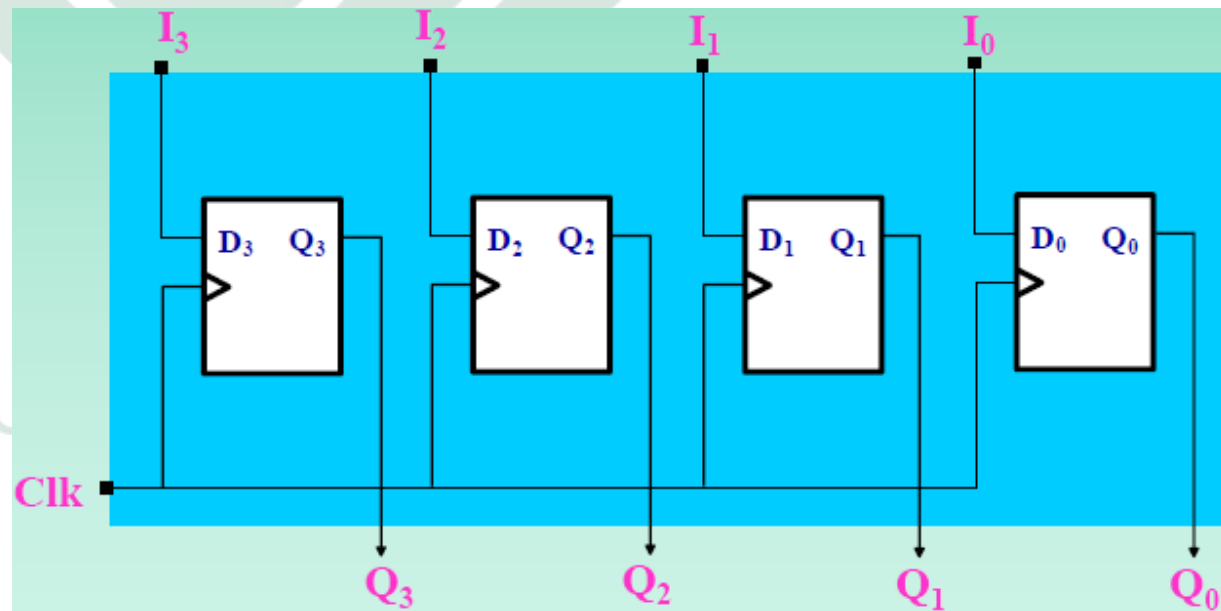
Registers

- Registers are bit wise extensions of flip-flops.
- Registers store one data word.

$$Q_i = I_i \text{ when Clk} = \uparrow$$



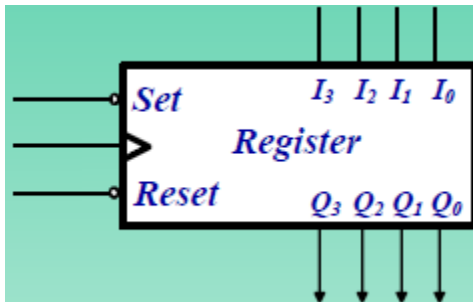
Graphic symbol



Register schematic

Registers with asynchronous set and reset

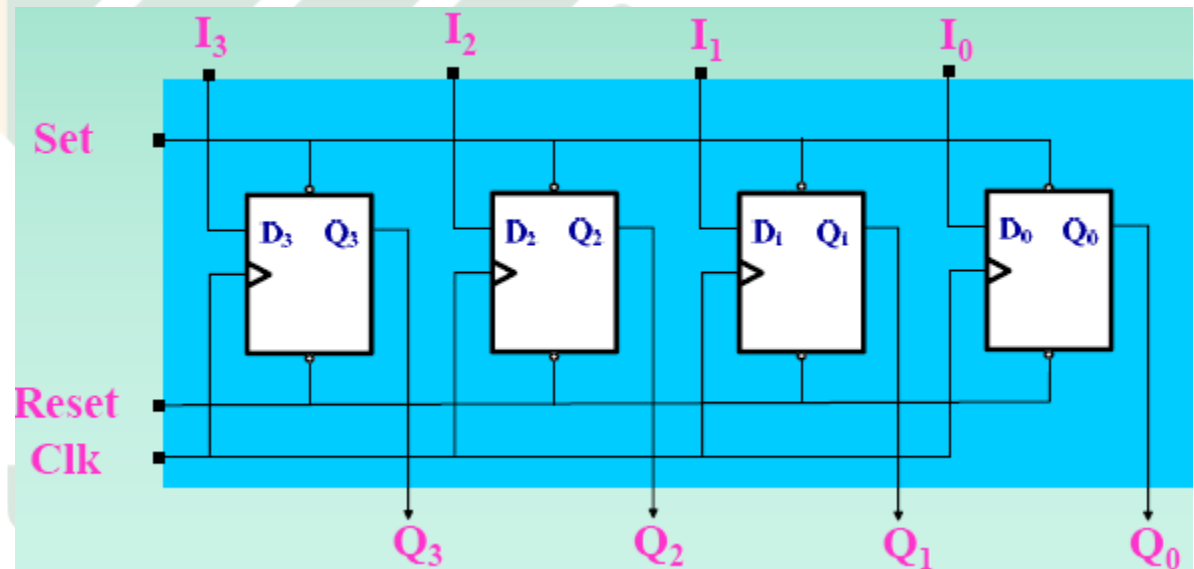
- Asynchronous setting and resetting is independent of clock signal.
- **Asynchronous** inputs are used to initialize the register.



Graphic symbol

S	R	Clk	Q_i
0	X	X	1
X	0	X	0
1	1	↑	I_i

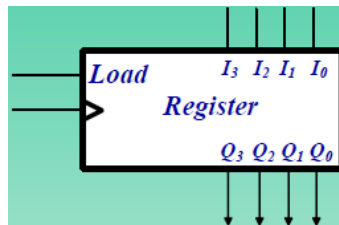
Operation table



Register schematic

Register with parallel load

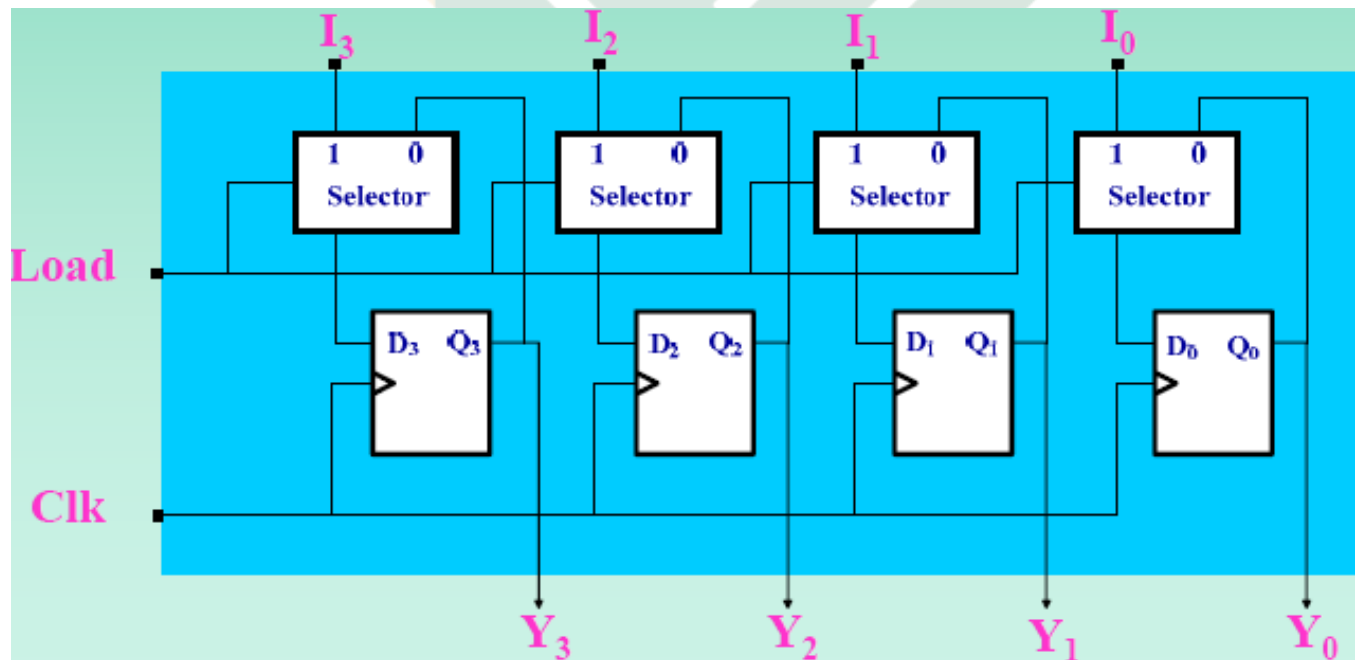
- Parallel load register can hold data indefinitely.
- It can also load new data when load signal is 1.



Graphic symbol

Present state	Next state
Load	$Q_3 Q_2 Q_1 Q_0$
0	No change
1	$I_3 I_2 I_1 I_0$

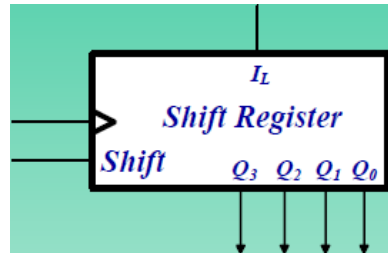
Operation table



Register schematic

Serial-in, parallel-out Shift register (shift-right)

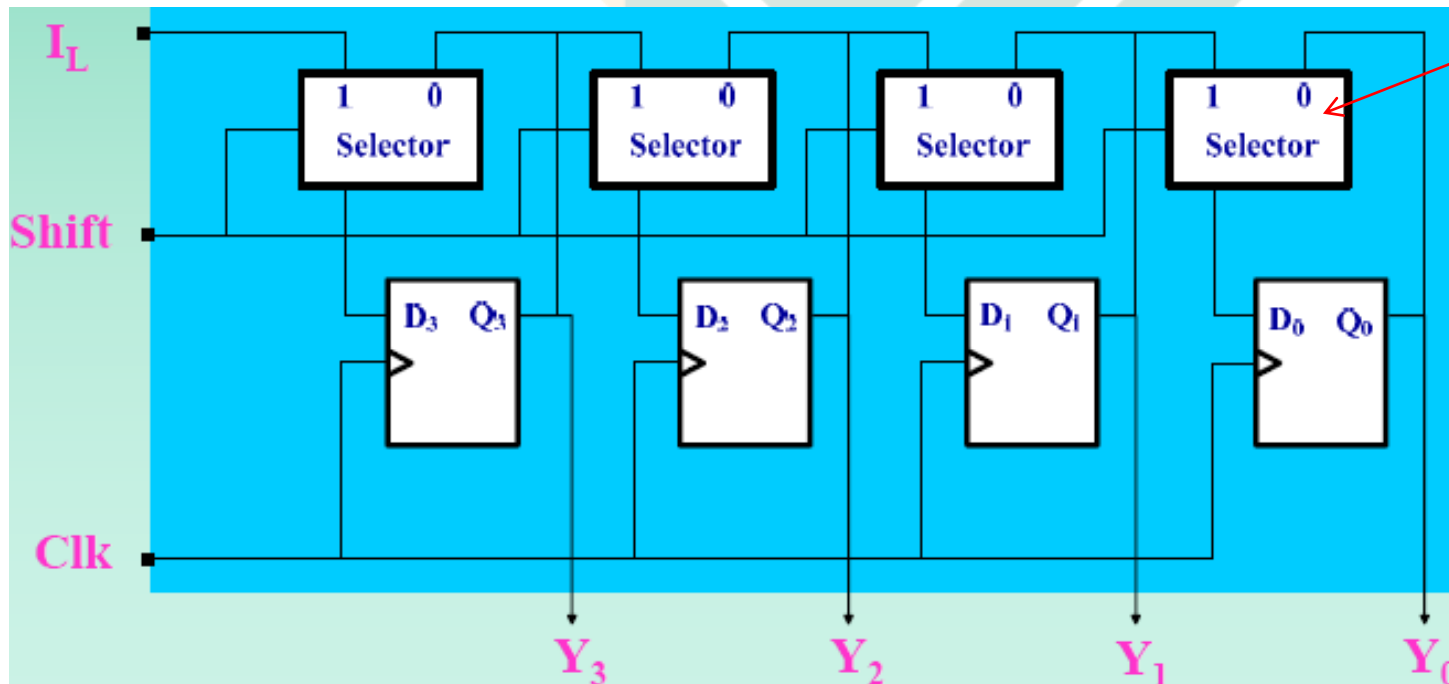
- Serial-in, parallel-out register converts serial data stream into parallel data stream.



Graphic symbol

Present state	Next state
Shift	$Q_3 Q_2 Q_1 Q_0$
0	No change
1	$I_L Q_3 Q_2 Q_1$

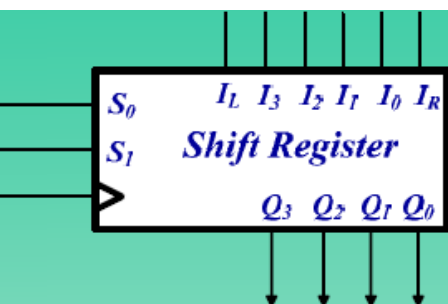
Operation table



Mux 2-to-1

Register schematic

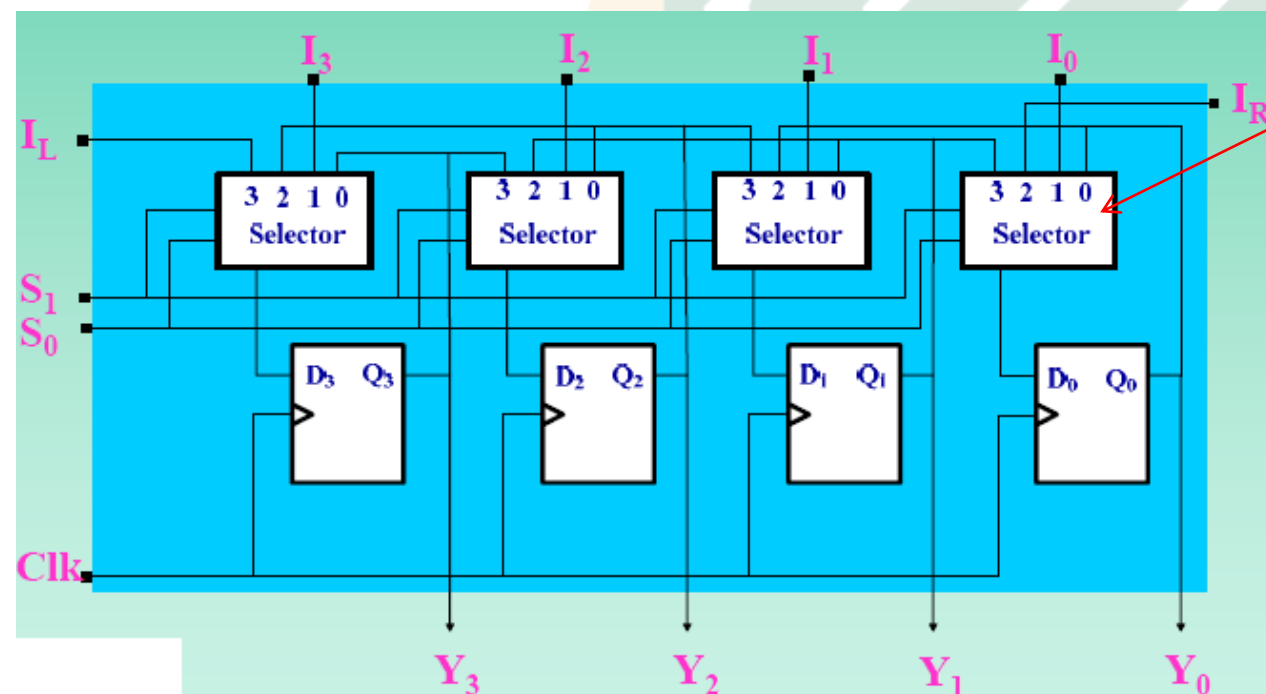
Shift register with parallel load (SRwPL)



Graphic symbol

Present state		Operation	Next state			
S_1	S_0		Q_3	Q_2	Q_1	Q_0
0	0	No change	Q_3	Q_2	Q_1	Q_0
0	1	Load input	I_3	I_2	I_1	I_0
1	0	Shift left	Q_2	Q_1	Q_0	I_R
1	1	Shift right	I_L	Q_3	Q_2	Q_1

Operation table



Register schematic

Mux 4-to-1

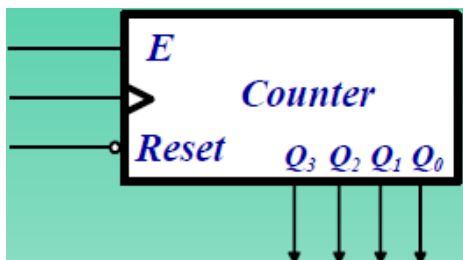
$$D_0 = S_1' S_0' Q_0 + S_1' S_0 I_0 + S_1 S_0' I_R + S_1 S_0 Q_1$$

$$D_i = S_1' S_0' Q_i + S_1' S_0 I_i + S_1 S_0' Q_{i-1} + S_1 S_0 Q_{i+1} \quad (1 \leq i \leq 2)$$

$$D_3 = S_1' S_0' Q_3 + S_1' S_0 I_3 + S_1 S_0' Q_2 + S_1 S_0 I_L$$

4-bit binary counter

- Counters increment (decrement) their content when enabled



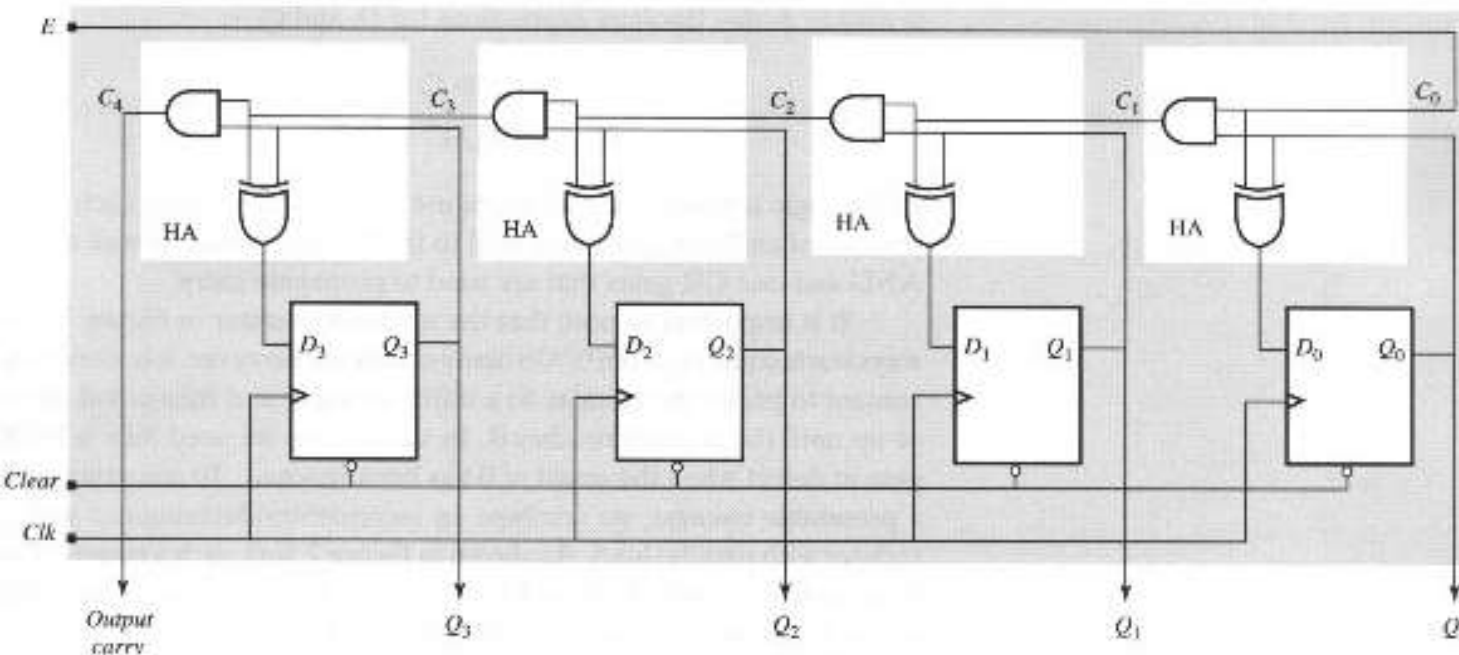
Graphic symbol

<i>E</i>	<i>Operations</i>
0	No change
1	Count

Operation table

Q_i	C_i	C_{i+1}	D_i
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

HA truth table



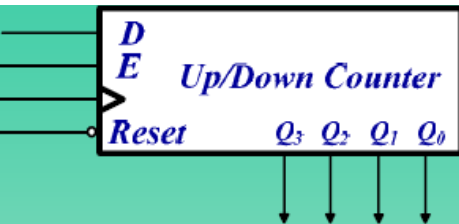
counter schematic

Up- or
Down-counter???

$$D_i = Q_i \oplus C_i$$

$$C_{i+1} = Q_i C_i$$

4-bit up/down binary counter



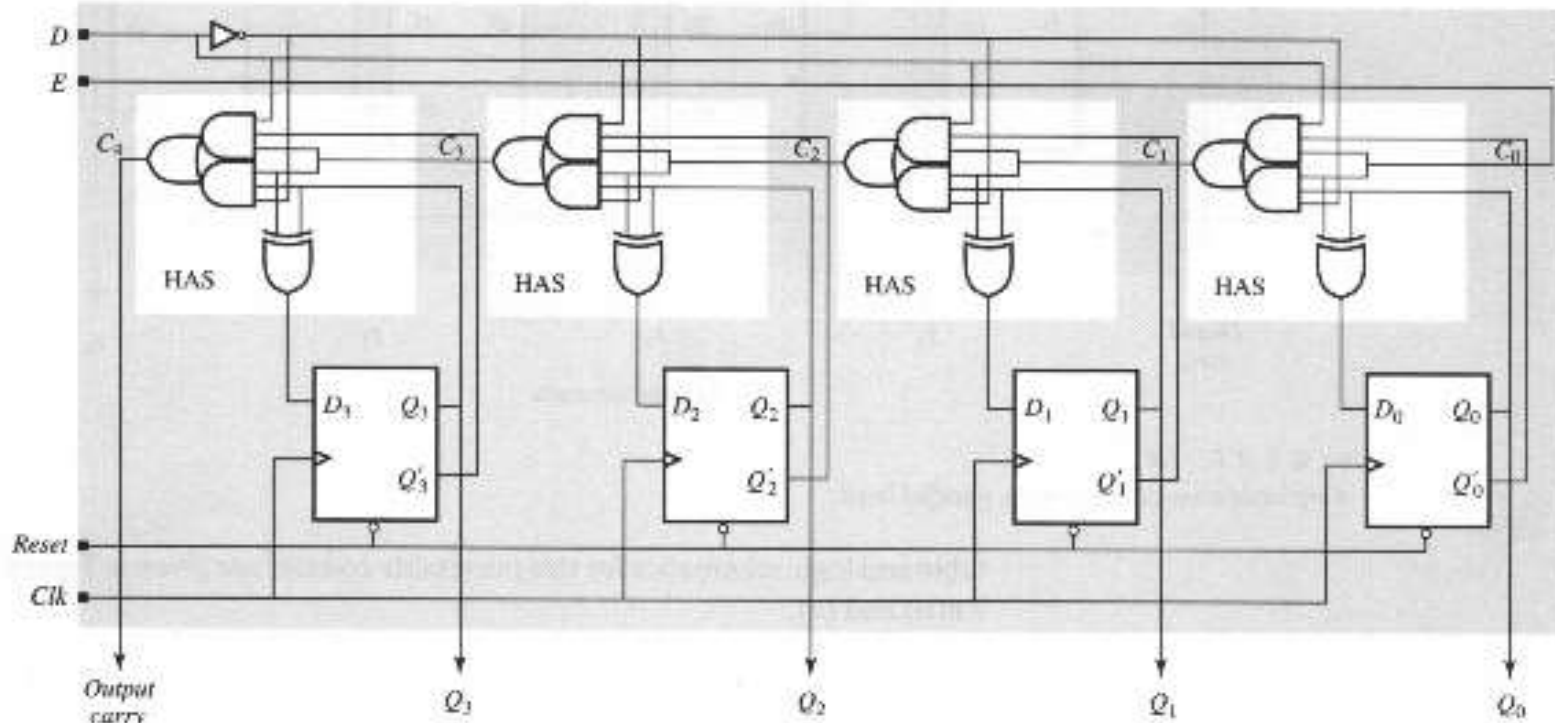
Graphic symbol

<i>E</i>	<i>D</i>	Operations
0	X	No change
1	0	Count up
1	1	Count down

Operation table

<i>E</i>	<i>D</i>	Q_i	C_i	C_{i+1}	D_i
1	0	0	0	0	0
1	0	0	1	0	1
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	1	1
1	1	1	0	0	1
1	1	1	1	0	0

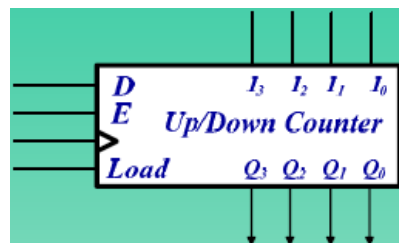
HAS truth table



Logic schematic

4-bit up/down binary counter with parallel load

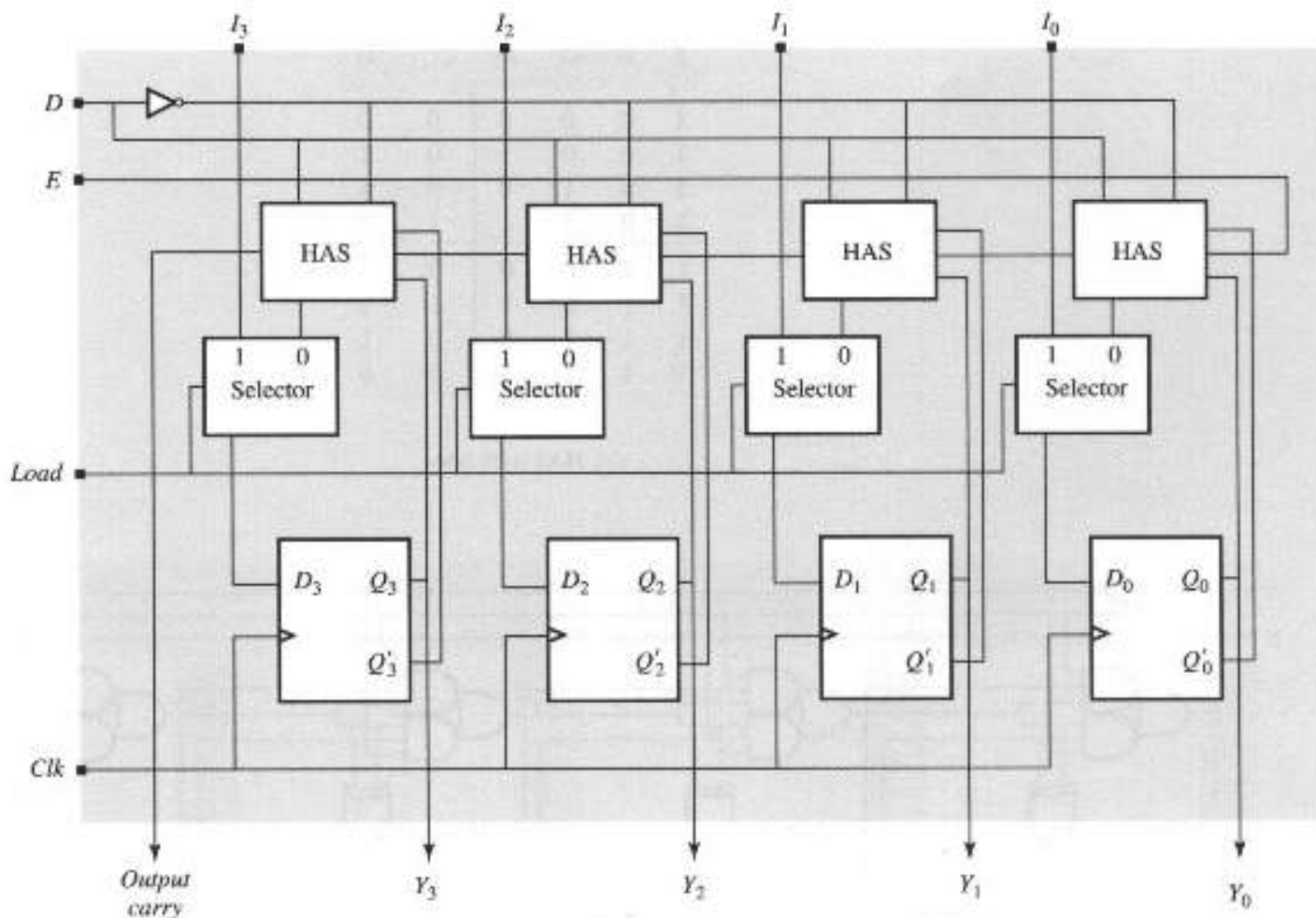
➤ This counter is sometimes called presettable counter



Graphic symbol

Load	E	D	Operations
0	0	X	No change
0	1	0	Count up
0	1	1	Count down
1	X	X	Load the input

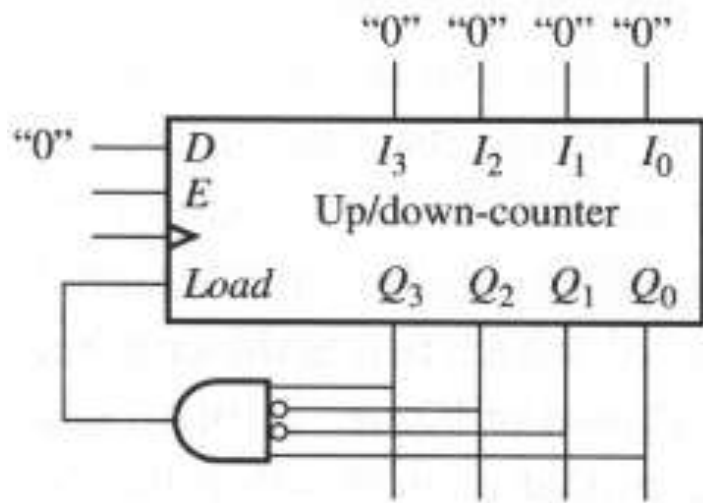
Operation table



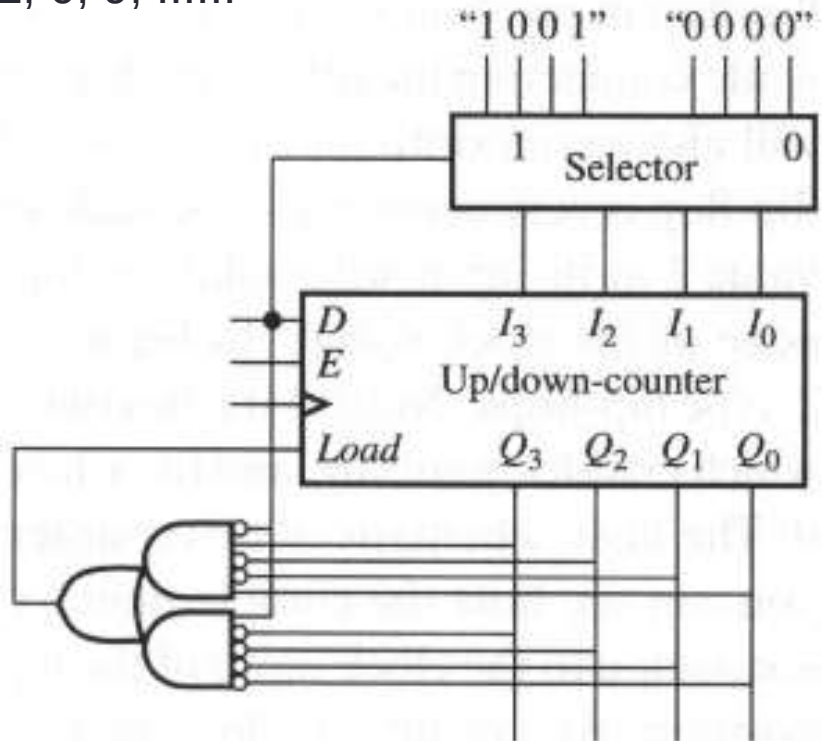
Logic schematic

BCD counters

- Up sequence: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0,
- Down sequence: 0, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0, 9,



BCD up-counter



BCD up/down-counter

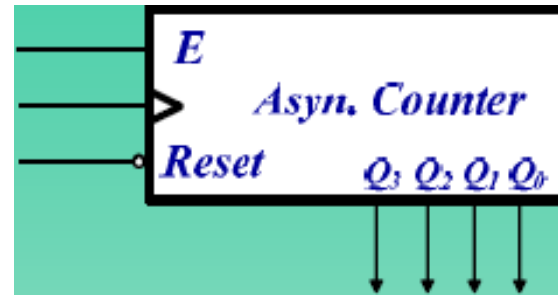
- Up counter loads 0 when counter content is 9 (1001)
- Up/down counter loads 0 when counter content is 9 (1001) and direction bit D=0
- Up/down counter loads 9 when counter content is 0 (0000) and direction bit D=1

Asynchronous counters

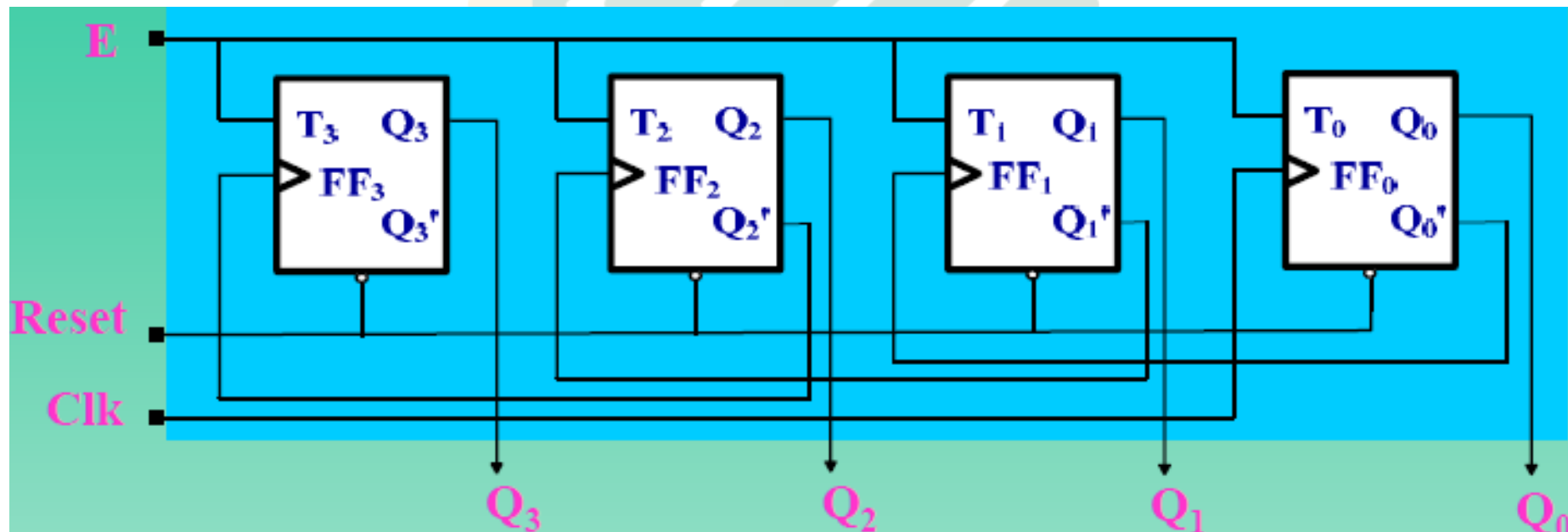
- Each FF in **synchronous counters** changes its output at the **same time**.
- FFs' output in **asynchronous counters** change values at **different times**.
- Advantage of asynchronous counters is **simple** and **low cost** (less gates).
- Weakness of asynchronous counters is **longer delays** in comparison with synchronous counters.

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4-bit asynchronous up counter

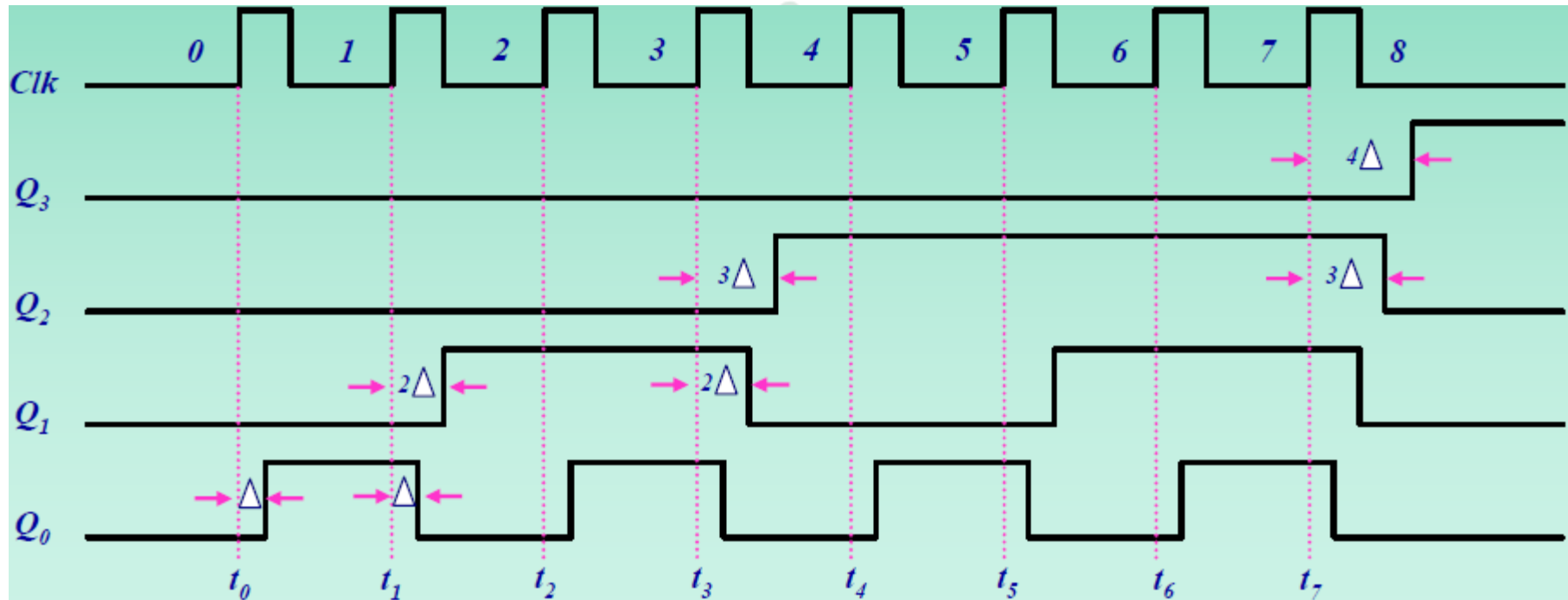


Graphic symbol



Logic schematic

4-bit asynchronous up counter

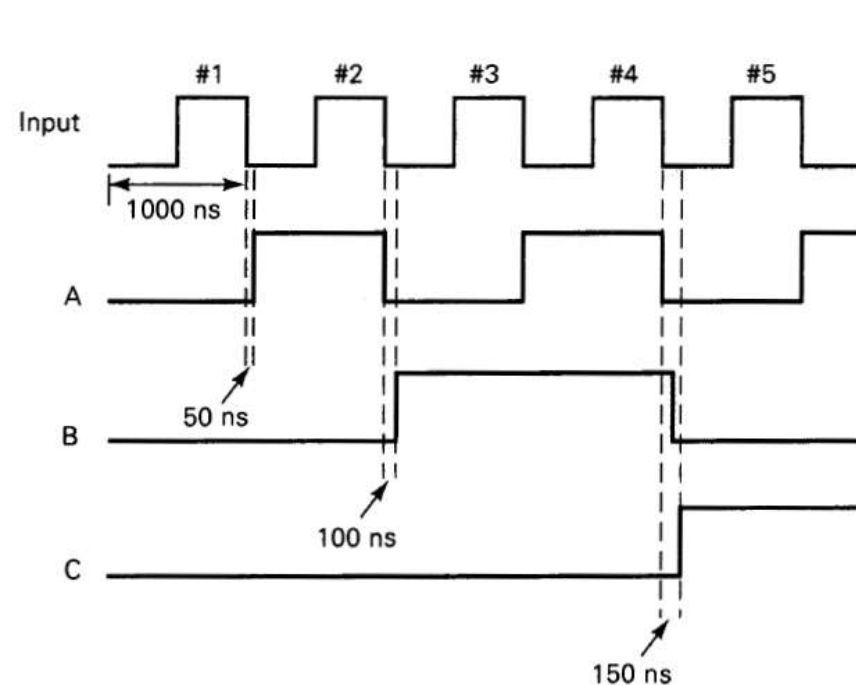


Timing diagram

Propagation Delay in Ripple Counters

- Ripple counters are simple—requiring the fewest components to produce a given operation
 - Cumulative propagation delay can cause problems at high frequencies
- The counter will perform accurately when the period of clock cycle must be longer than the total propagation delay of it.
 - For proper operation: $T_{\text{clock}} \geq N \times t_{\text{pd}}$
 - T_{clock} : the period of clock cycle
 - N : the number of FFs
 - T_{pd} : the propagation delay of a FF (Δ)
 - Maximum frequency: $F_{\text{max}} = 1 / (N \times t_{\text{pd}})$

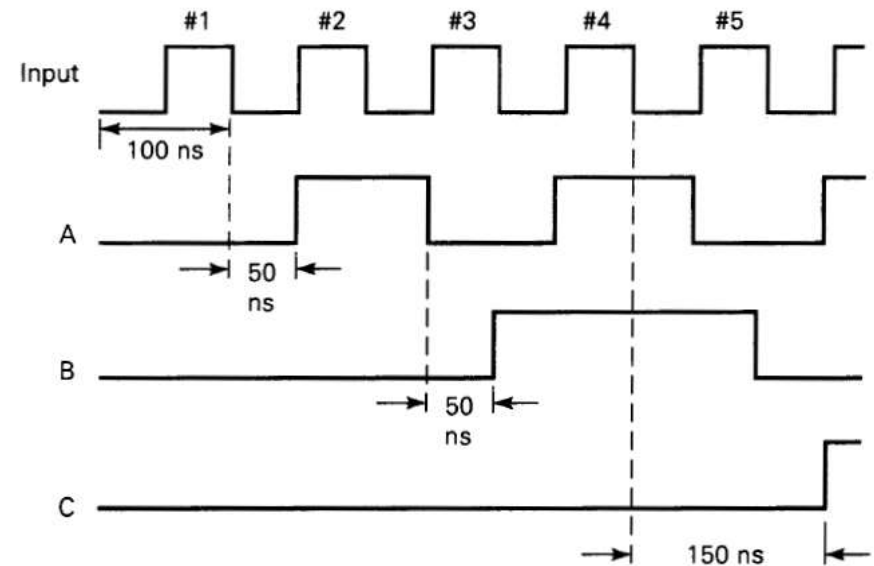
Propagation Delay in Ripple Counters



- $T = 1000 \text{ ns}$
- $t_{pd} = 50 \text{ ns}$

→ $T \geq 3 \times t_{pd}$

Counter operates properly



- $T = 100 \text{ ns}$
- $t_{pd} = 50 \text{ ns}$

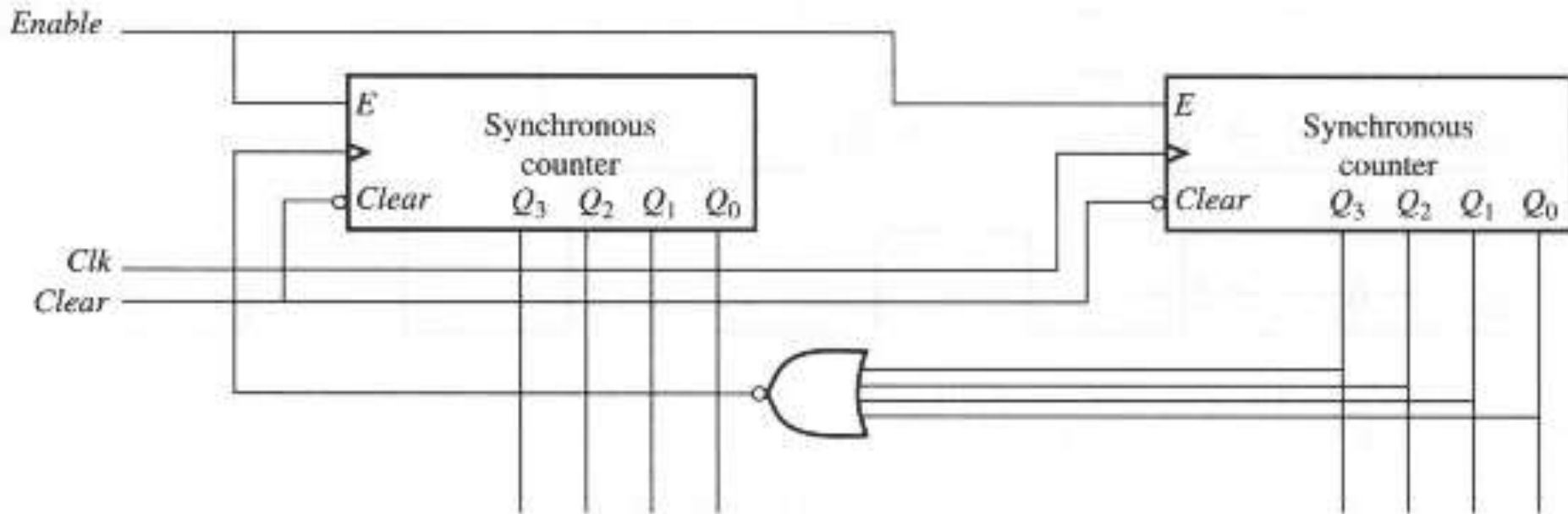
→ $T < 3 \times t_{pd}$

Counter operates wrongly

The CBA = 100
does not occur

8-bit mixed-mode up counter

- Mixed-mode counter consists of:
 - (1) asynchronous counters connected synchronous slices.

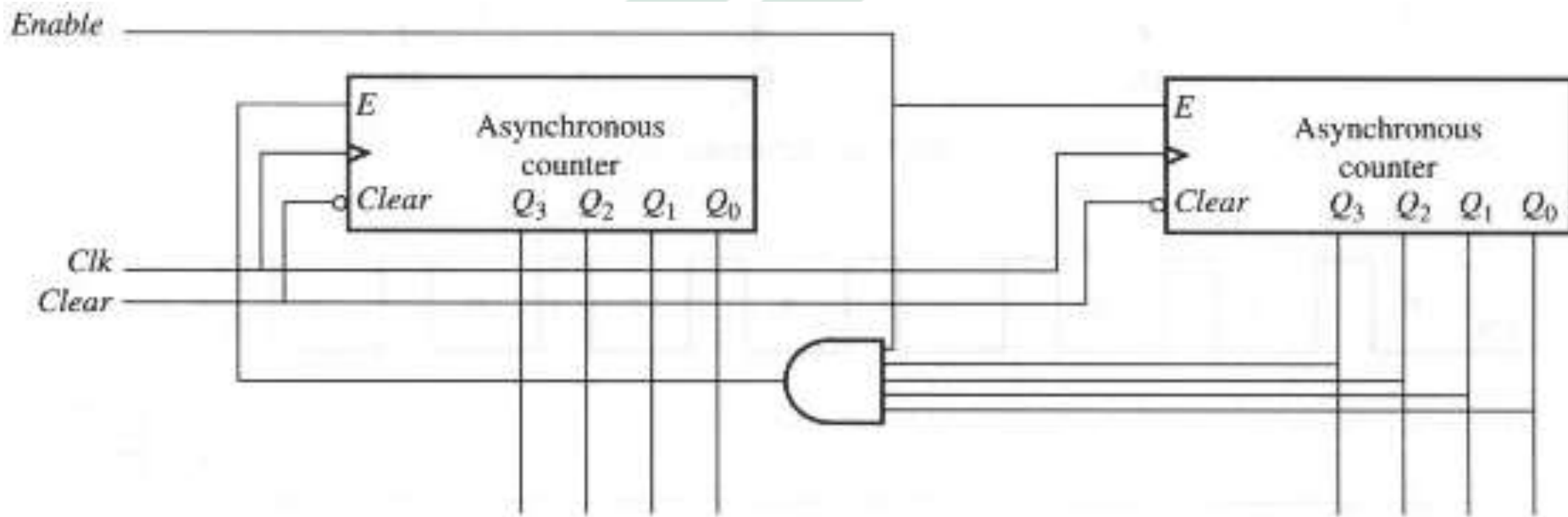


Asynchronous counter with 4-bit synchronous slices

8-bit mixed-mode up counter (cont.)

➤ Mixed-mode counter consists of:

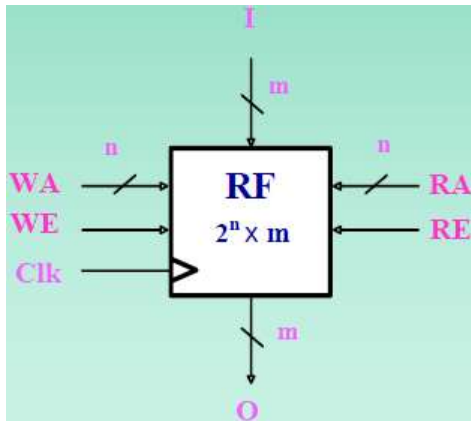
- (2) synchronous counters connected asynchronously.



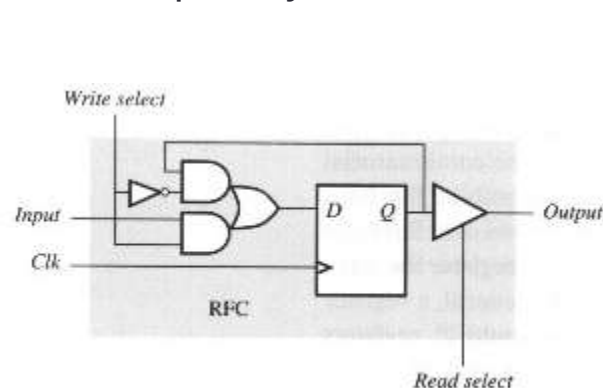
Synchronous counter with 4-bit asynchronous slices

Register-file

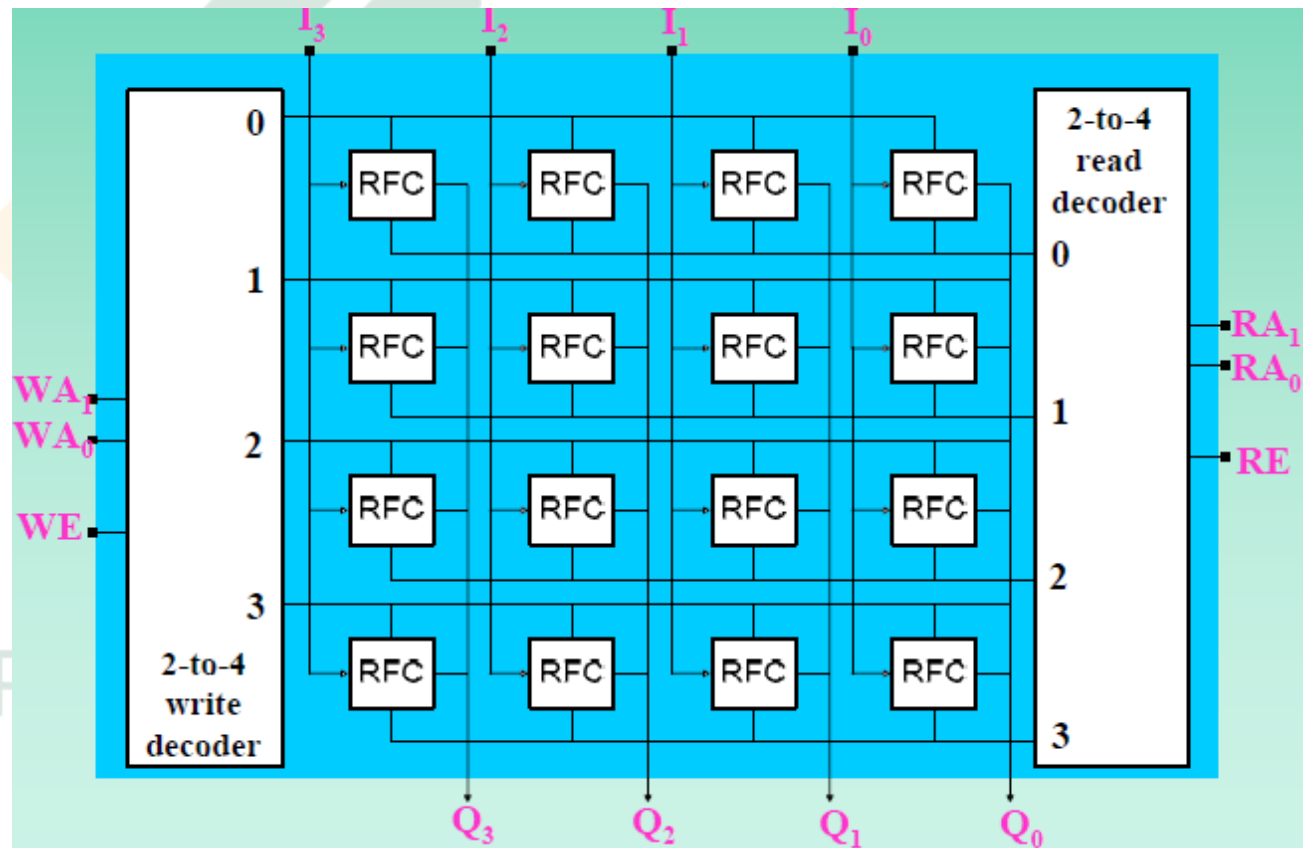
- Register-file is used as fast temporary storage



Graphic symbol



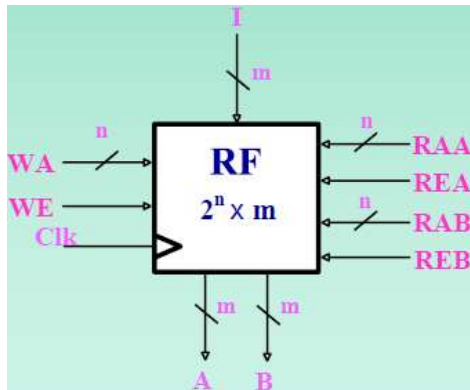
Register-file cell



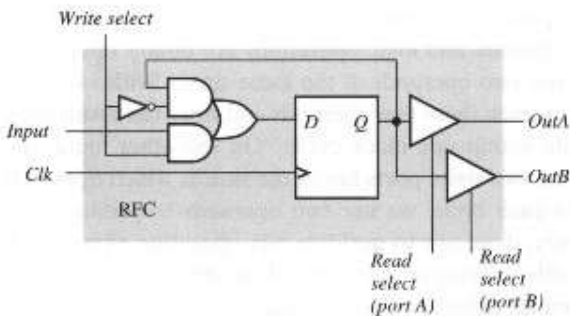
Logic schematic

Register-file with 1 write port and 2 read ports

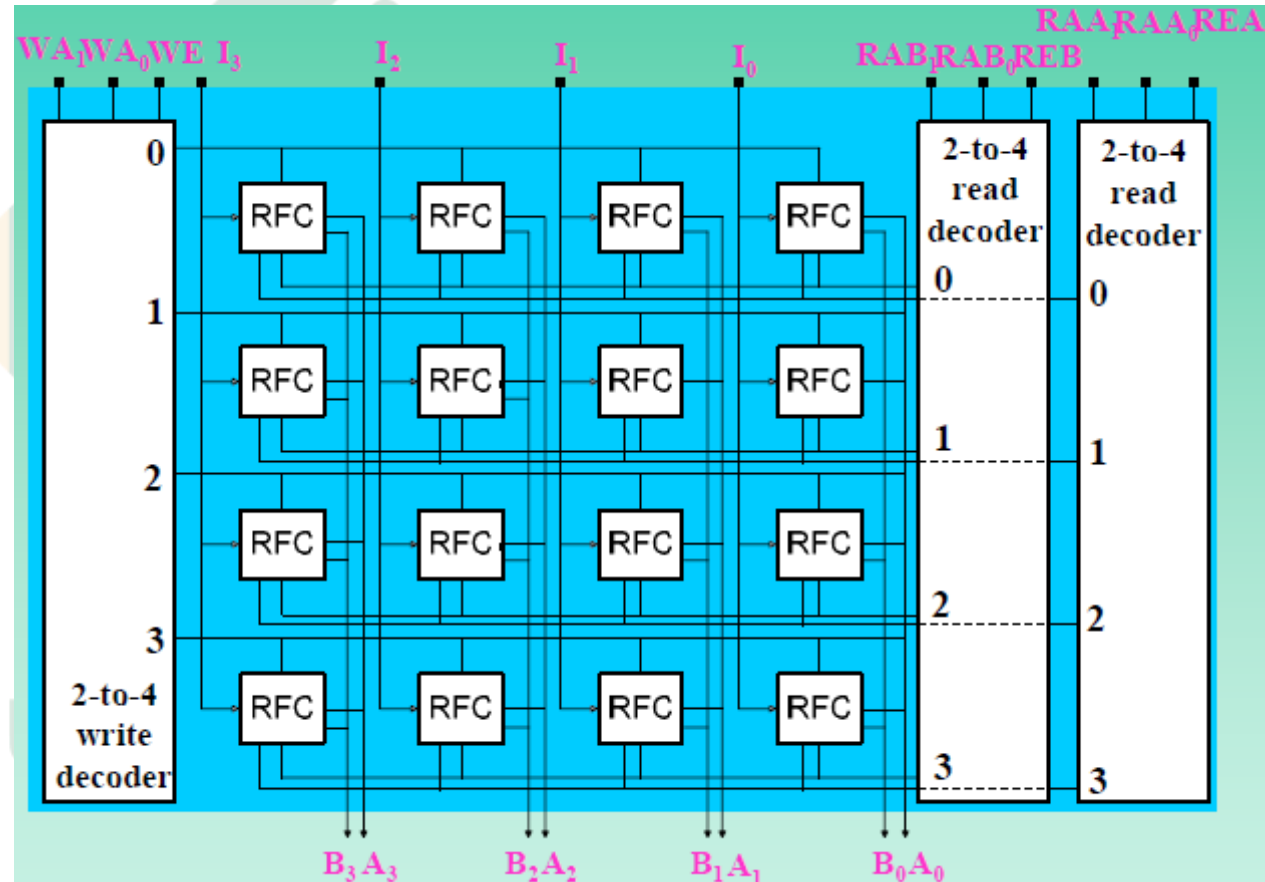
- This register-file is used for reading two operands and writing one result in each clock cycle.



Graphic symbol



Register-file cell

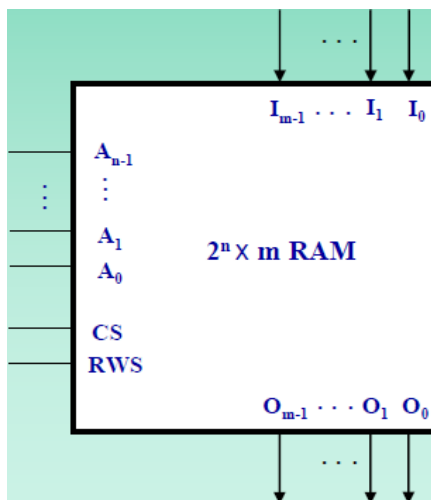


Logic schematic

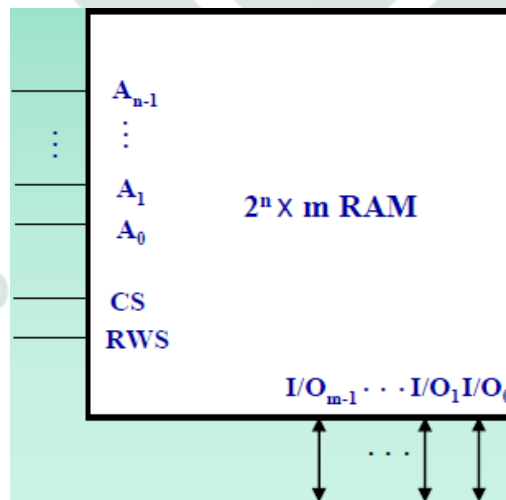
Random Access Memory (RAM)

MEMORY ADDRESS		MEMORY CONTENT
Binary	Decimal	
0...000	0	011...0100
0...001	1	011...0100
0...010	2	101...1100
0...011	3	101...0001
0...100	4	011...0101
0...101	5	010...0101
0...110	6	110...0011
0...111	7	101...0001
⋮	⋮	⋮
1...110	2^n-2	000...0010
1...111	2^n-1	111...0110

Memory address
and content



Graphic symbol



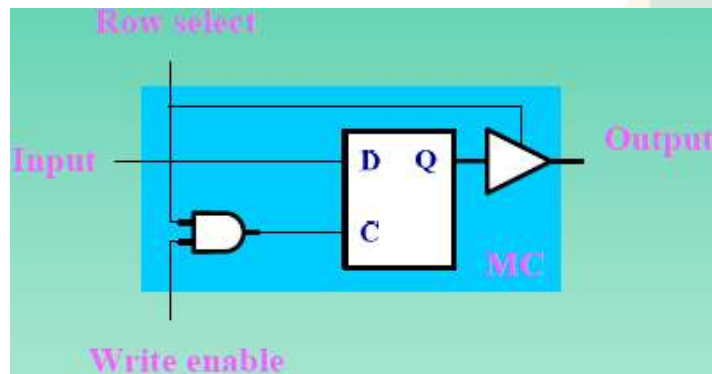
Graphic symbol

CS: Chip Select

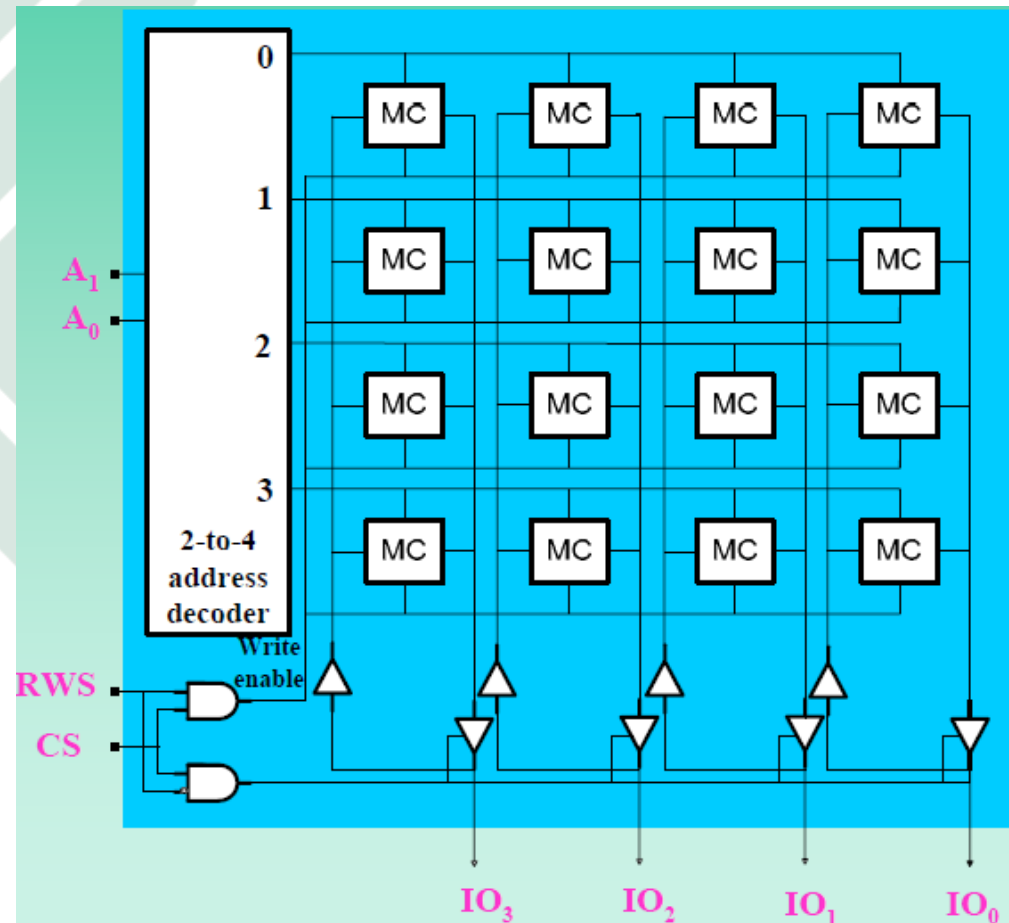
RWS: Read/Write Select
0: Read, 1: Write

RAM organization

- Ram memory cells can be static or dynamic.
- Static RAM's do not lose data with time.
- Dynamic RAM's must be refreshed.

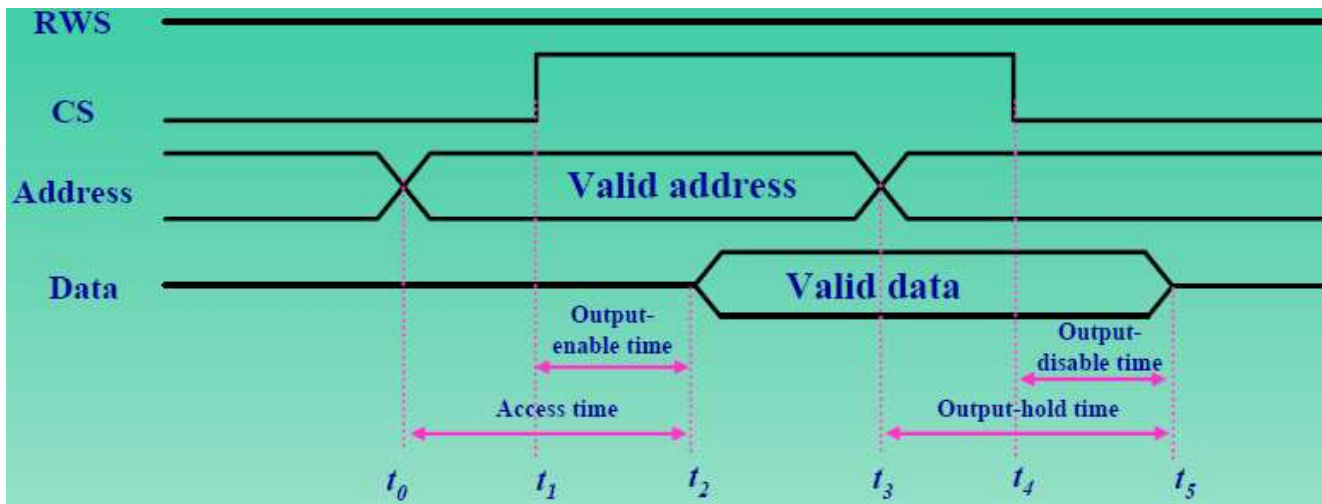


Memory cell

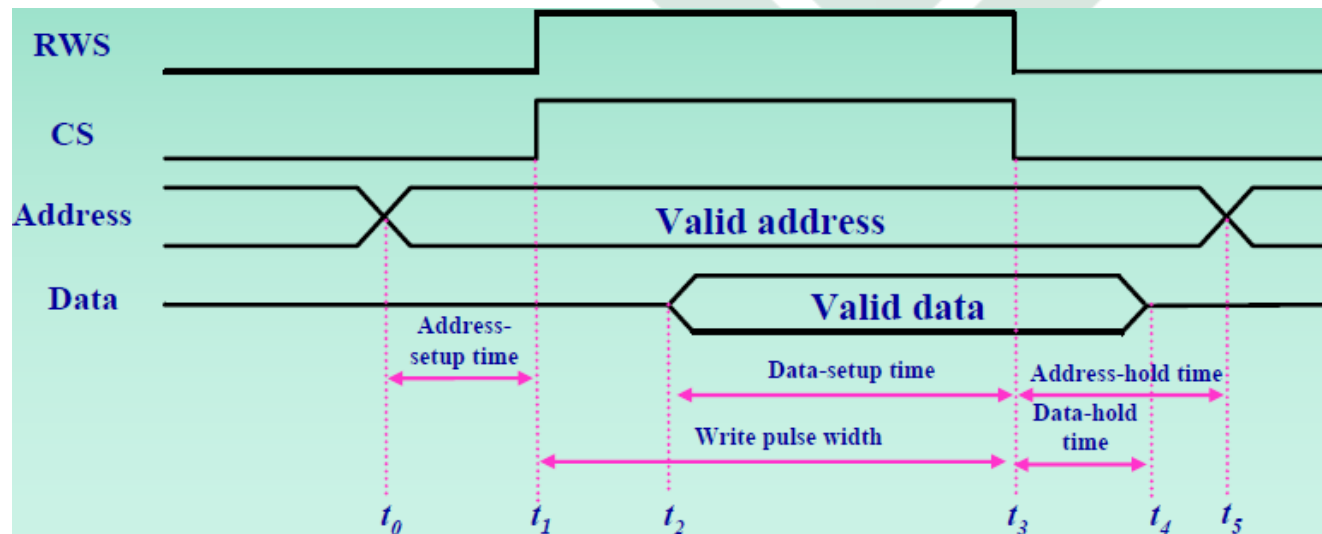


Memory schematic

RAM timing

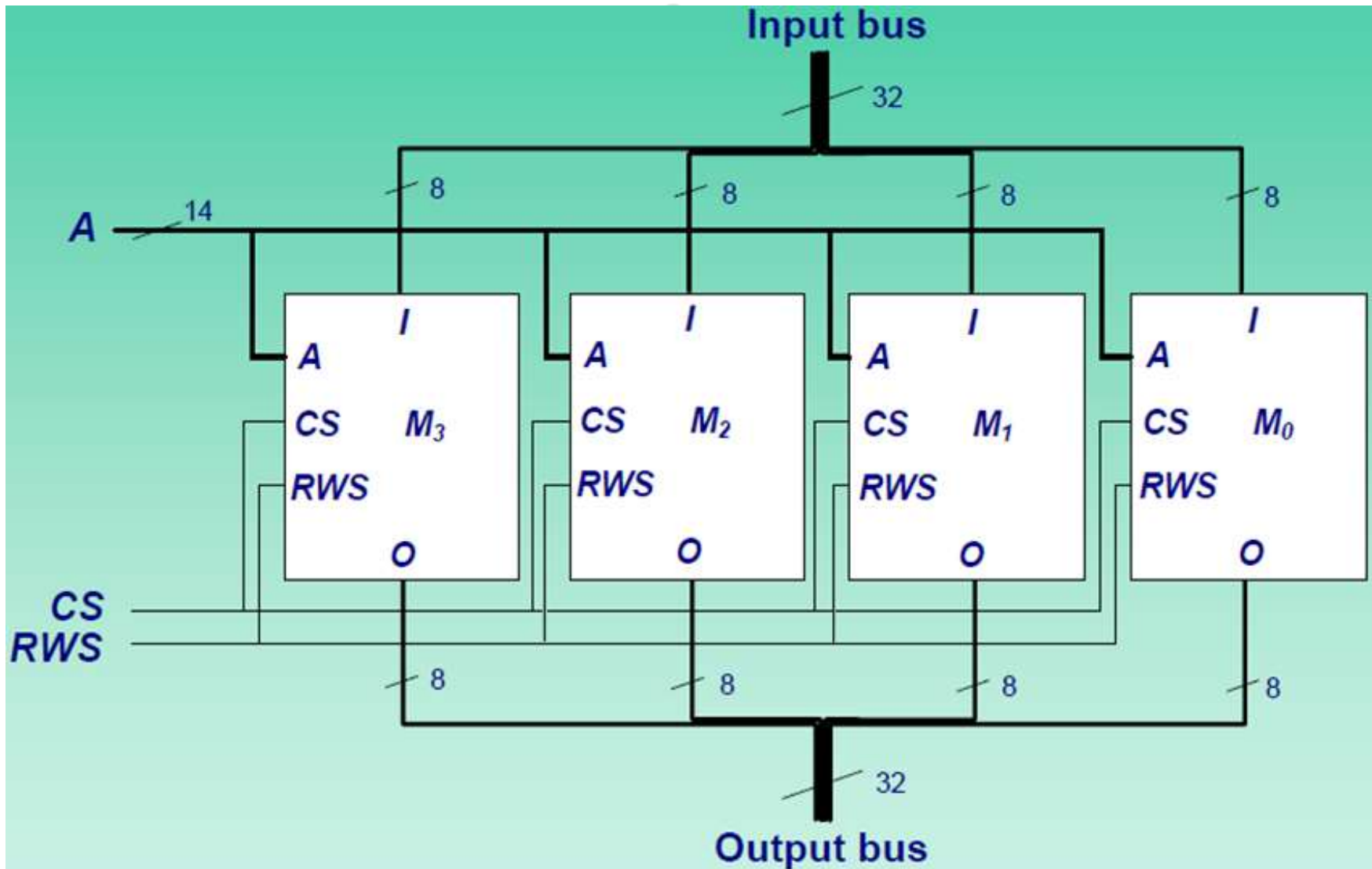


Read-cycle timing

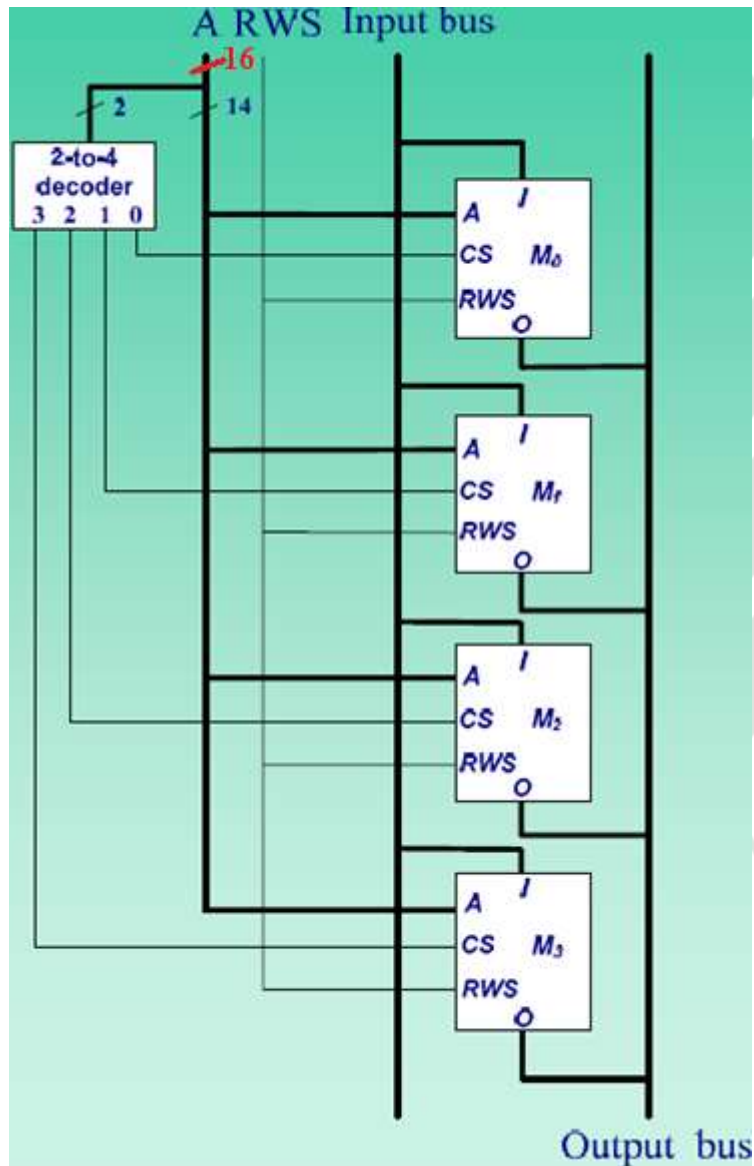


Write-cycle timing

16K x 32 RAM design with 16K x 8 RAMs (to obtain wider bit widths)



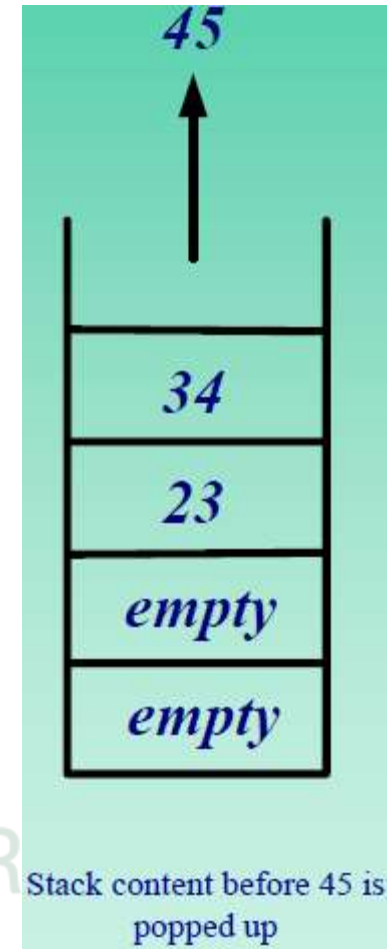
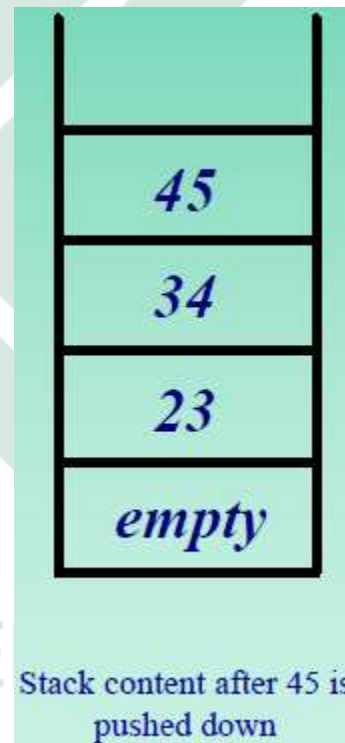
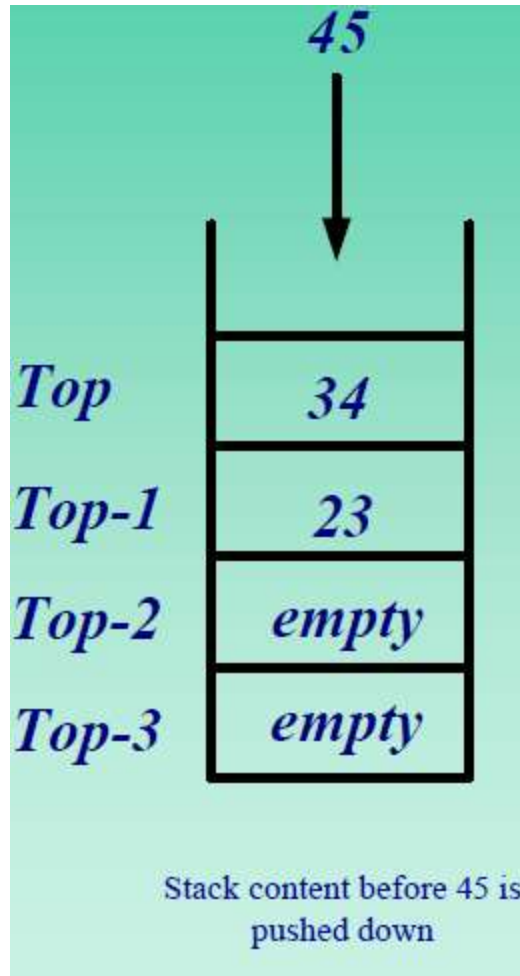
64K x 8 RAM design with 16K x 8 RAMs (to obtain a larger memory)



Addresses

0
 $\vdots$
 $2^{14}-1$
 2^{14}
 $\vdots$
 $2^{15}-1$
 2^{15}
 $\vdots$
 $2^{15}+2^{14}-1$
 $2^{15}+2^{14}$
 $\vdots$
 $2^{16}-1$

Push-down stack principle



4-Word push-down stack

Push/Pop	Enable	Operations
X	0	No change
0	1	Push
1	1	Pop

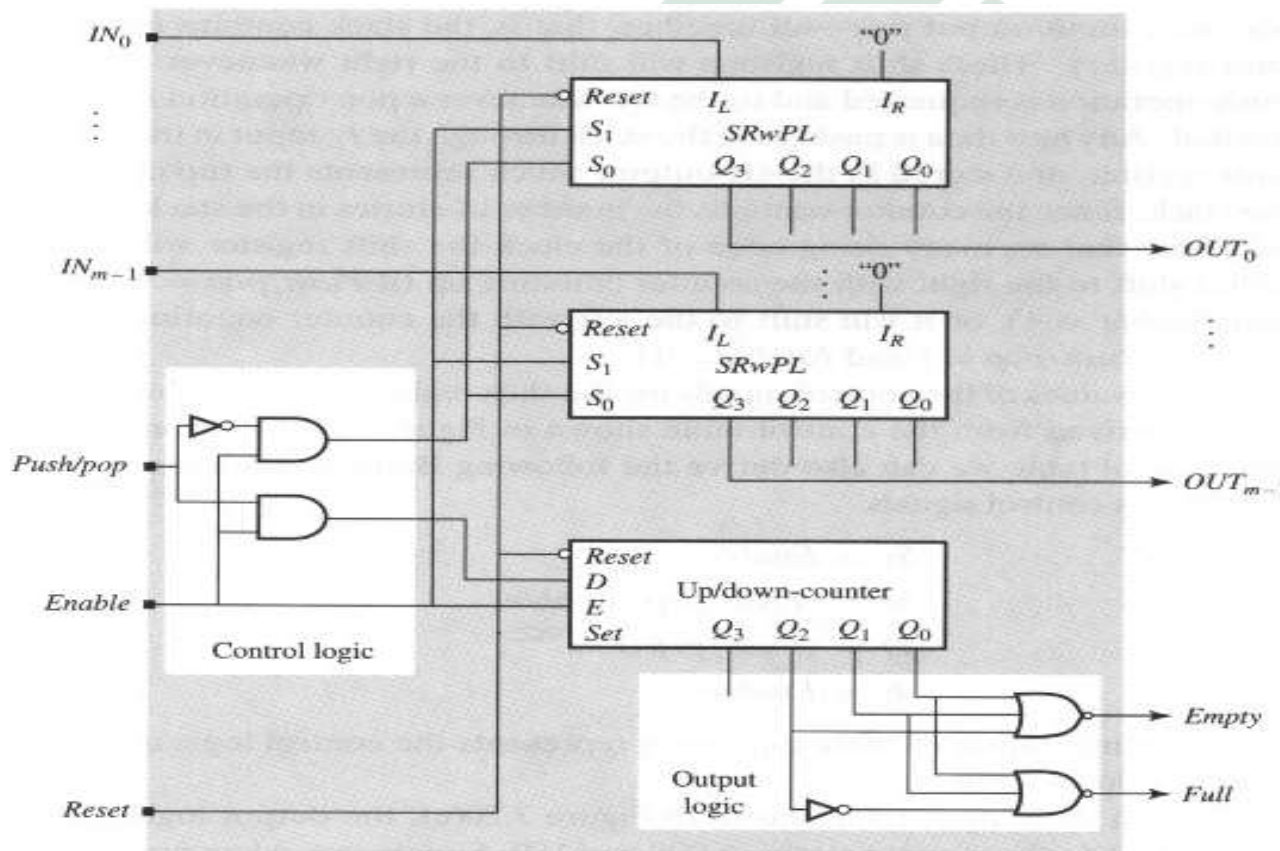
Operation table

		Shift register controls		Counter controls	
Push/Pop Enable		S ₁	S ₀	D	E
X	0	0	0	X	0
0	1	1	1	0	1
1	1	1	0	1	1

Control table

Counter outputs			Empty	Full
Q ₂	Q ₁	Q ₀		
0	0	0	1	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	1

Output table



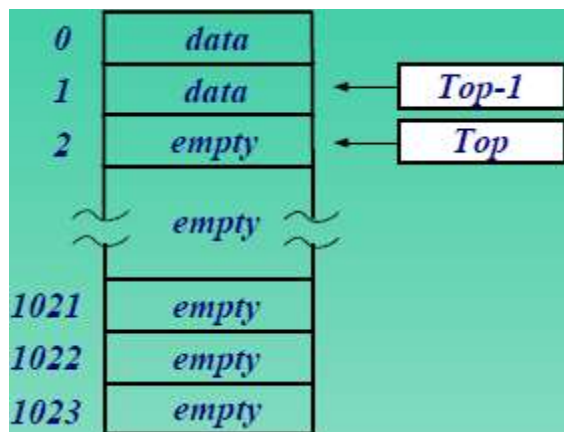
Stack schematic

Push=Shift right
Pop=Shift left

RING

Sửa trong sách
chỗ tín hiệu Set
của Counter

Push-down stack implemented with a 1K RAM



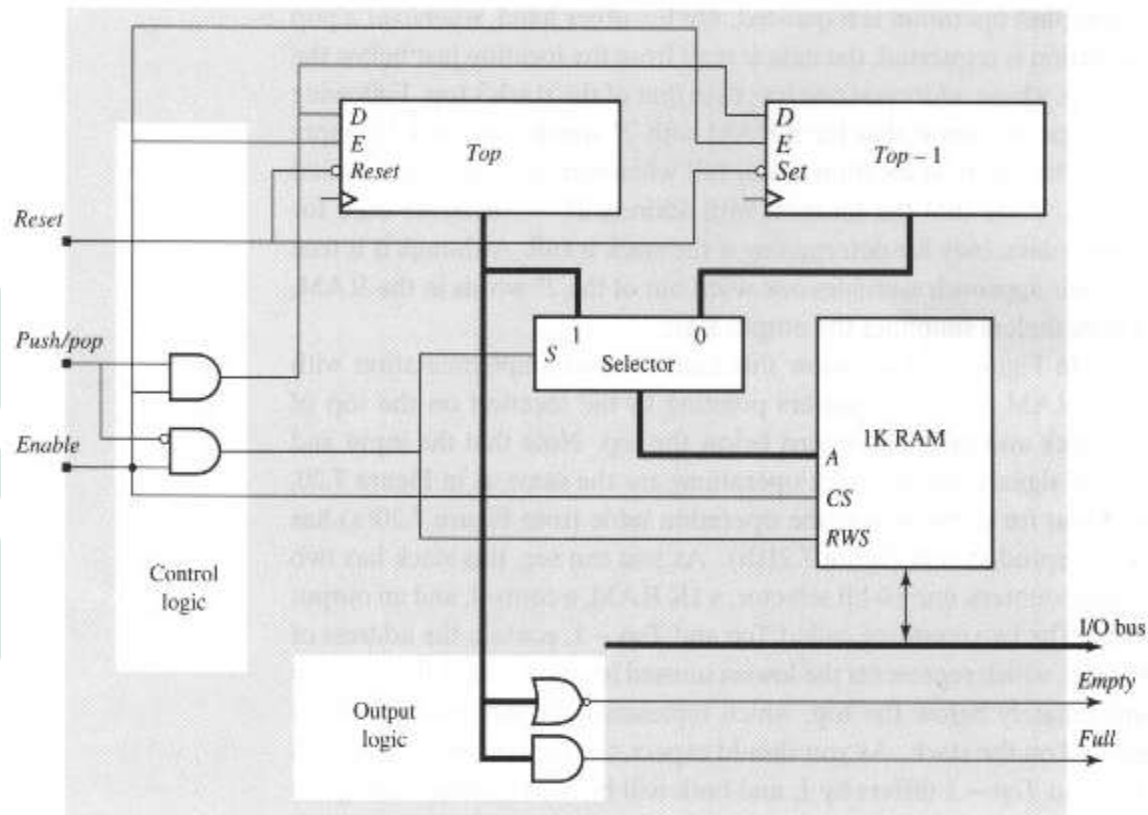
Symbolic design

Push/Pop	Enable	Operations
X	0	No change
0	1	Push
1	1	Pop

Operation table

Push/Pop	Enable	Selector control		Memory controls		Counter controls	
		S		CS	RWS	D	E
X	0	X		0	0	X	0
0	1	1		1	1	0	1
1	1	0		1	0	1	1

Control table



Push: Data $\rightarrow$ RAM (*TOP*); Increment Top, Top-1

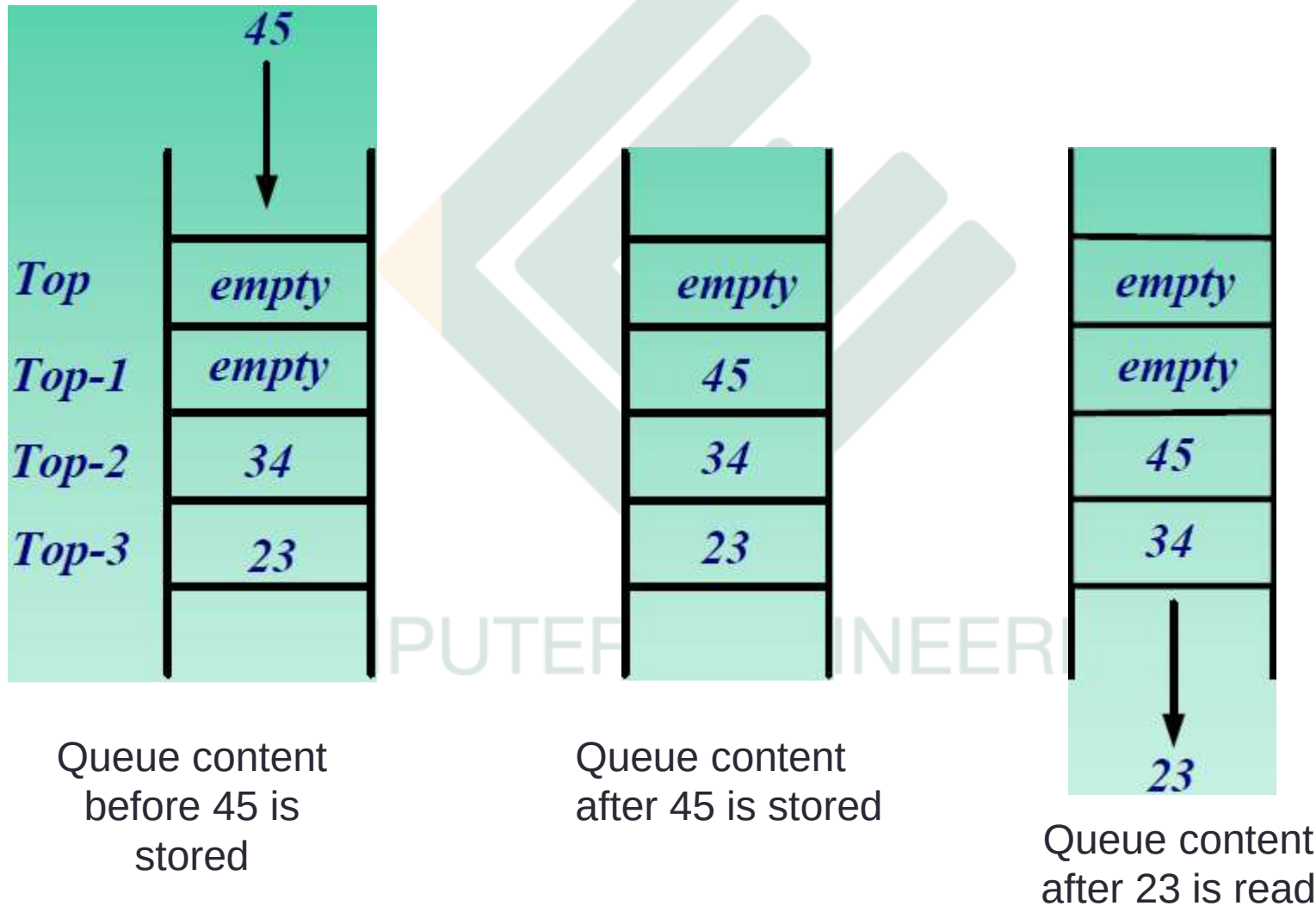
Pop: RAM (*Top-1*) $\rightarrow$ Data; Decrement Top, Top-1

Stack is full when Top=1023

Stack is empty when Top=0

Location with address 1023 is never loaded

FIFO queue principles



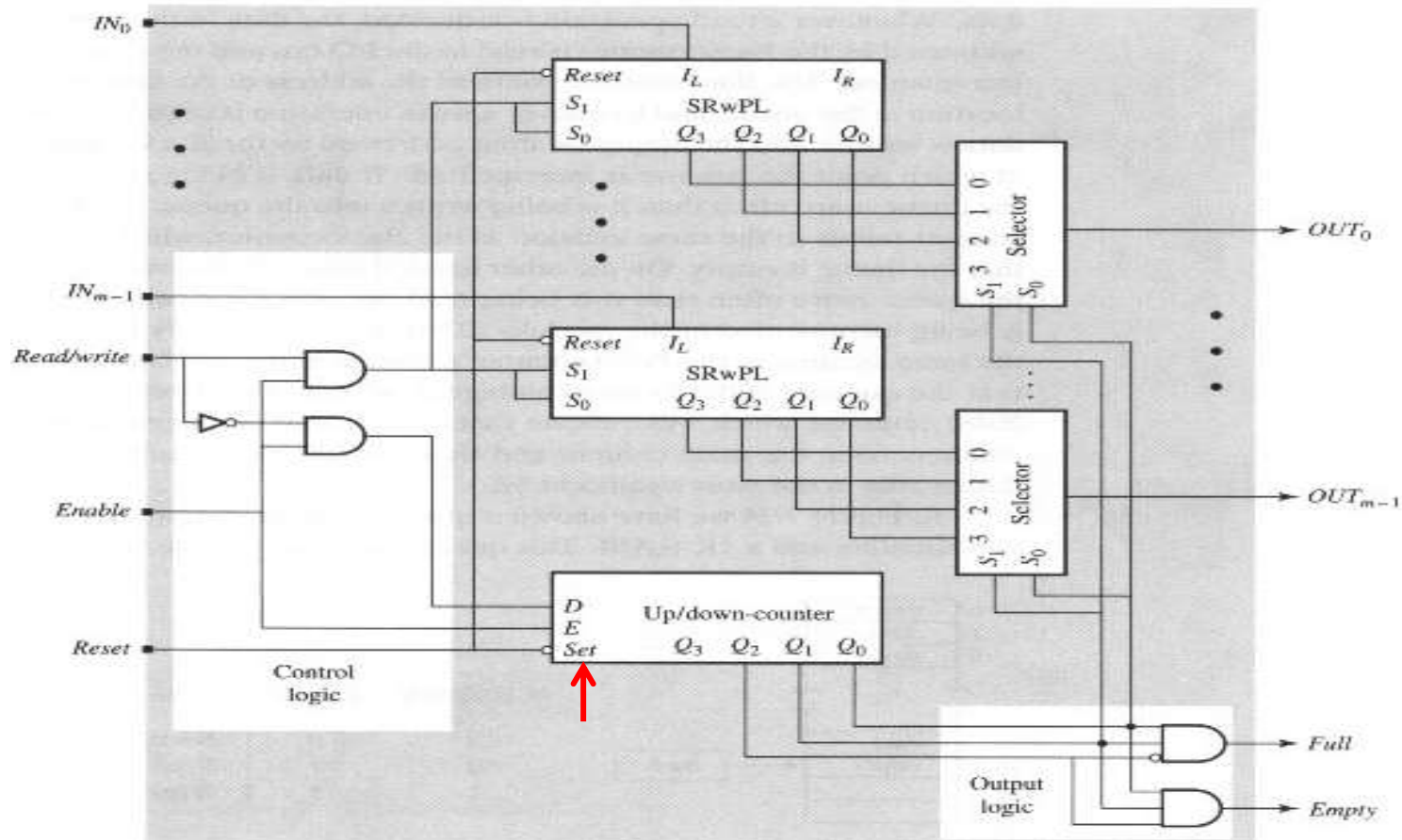
4-word FIFO queue

READ/WRITE	ENABLE	OPERATIONS
X	0	No change
0	1	Read
1	1	Write

Operation table

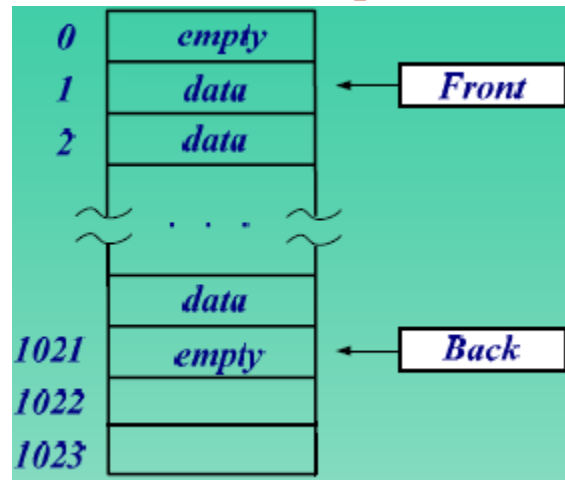
READ/WRITE	ENABLE	S_1	S_0	D	E
X	0	0	0	X	0
0	1	0	0	1	1
1	1	1	1	0	1

Control table



Queue schematic

FIFO queue implemented with a 1K RAM



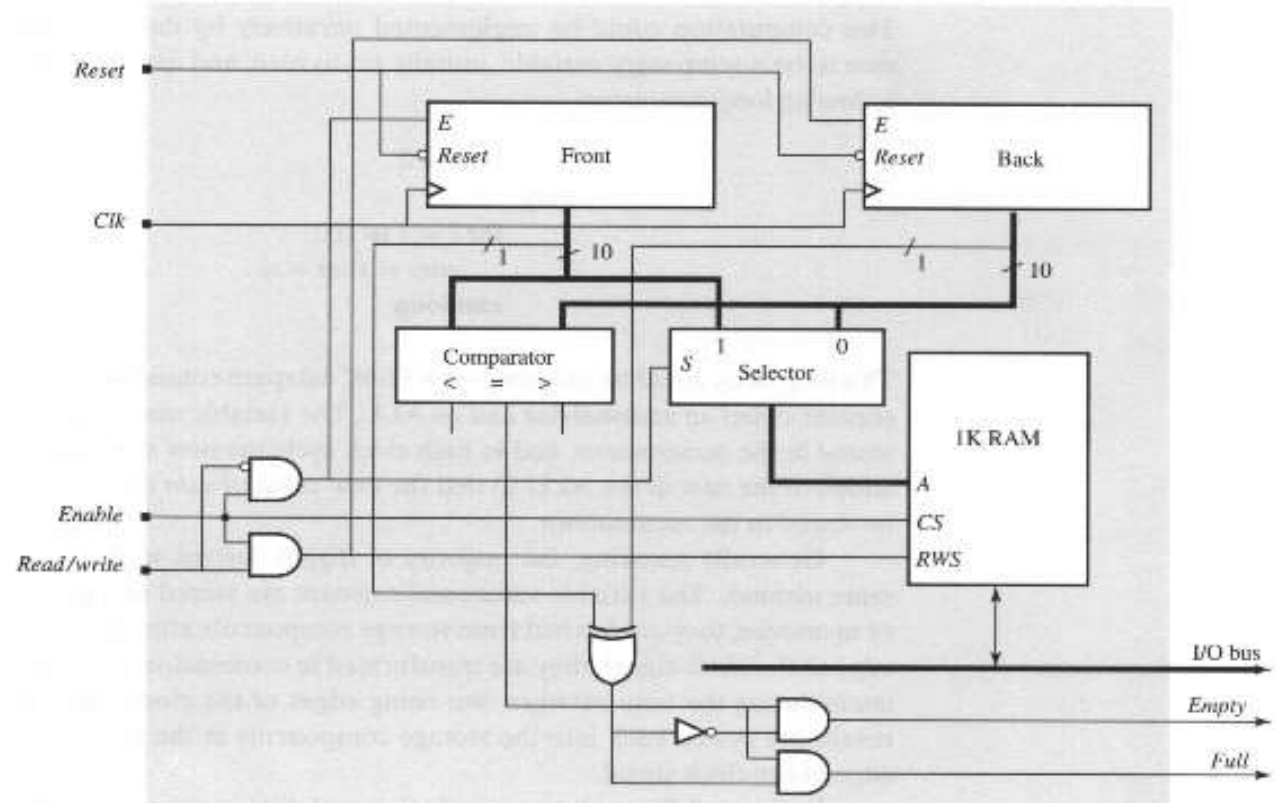
Symbolic design

Read/Write	Enable	Operations
X	0	No change
0	1	Read
1	1	Write

Operation table

Read/Write Enable		S	CS	RWS	E (Front)	E (Back)
X	0	X	0	X	0	0
0	1	1	1	0	1	0
1	1	0	1	1	0	1

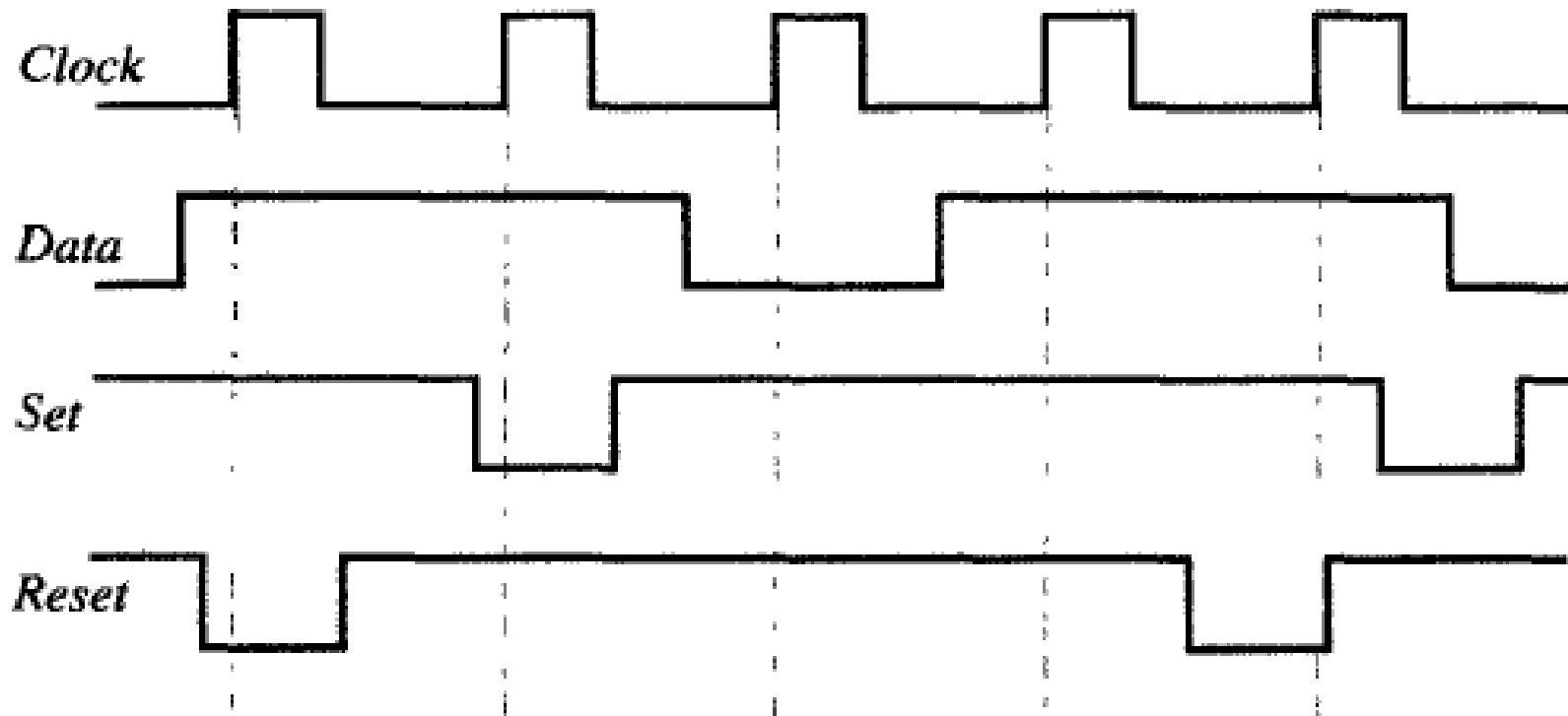
Control table



Schematic

Bài tập – Thanh ghi

- Cho giản đồ xung sau, vẽ giản đồ xung output của FF - D



Bài tập – Thanh ghi & Thanh Ghi dịch

- Thiết kế 1 thanh ghi 4 bit có 2 tín hiệu nạp dùng để điều khiển việc nạp dữ liệu từ 2 nguồn khác nhau
- Thiết kế thanh ghi 16 bit có khả năng làm 2 chức năng sau:
 - Nạp dữ liệu mới
 - Hoán đổi byte trọng số thấp và byte trong số cao (8 bit thấp lên 8 bit cao).

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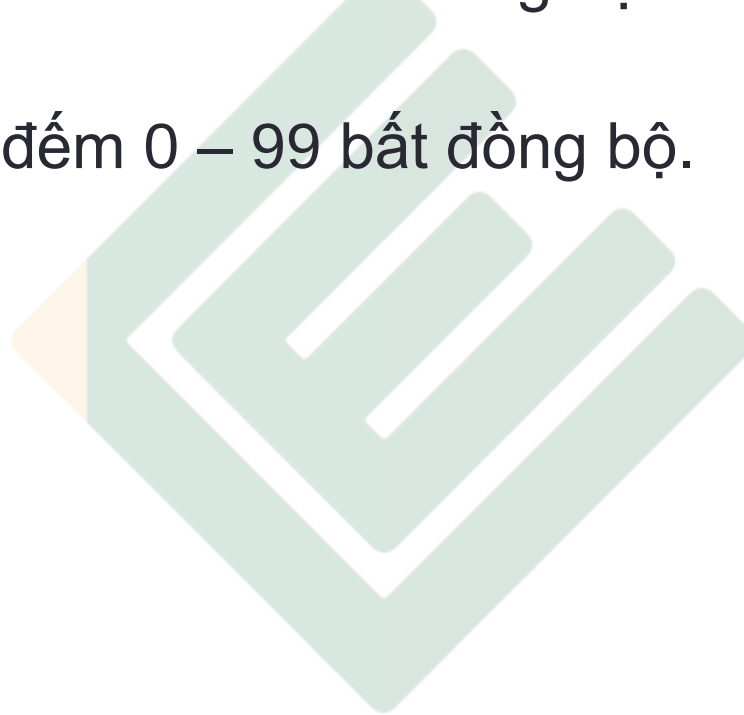
Bài tập – Bộ Đếm

- Thiết kế mạch đếm nhị phân chỉ đếm lên:
 - Số chẵn: 0, 2, 4, 6, 8, 0,
 - Số lẻ: 1, 3, 5, 7, 1, ...
- Thiết kế các cổng luận lý sao cho mạch đếm đếm đến các giá trị sau:
 - 7
 - 14
 - 15
- Thiết kế mạch đếm chỉ đếm các số sau: 1, 5, 9, 11, 13, 15, 1,

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Bài tập –Bộ Đếm

- Thiết kế mạch đếm 0 – 9 bất đồng bộ 4 bit sử dụng:
 - FF J-K.
- Thiết kế mạch đếm 0 – 99 bất đồng bộ.



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Bài tập – Thiết kế bộ nhớ

➤ Thiết kế bộ nhớ:

- 256K × 8 RAM sử dụng thành RAM 256K × 1 RAM
- 256K × 8 RAM sử dụng thành RAM 64K × 8 RAM



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